



SENDER
NIGHT
MADRID

tvOS vs iOS

About tvOS

tvOS is the operating system used in Apple TV.

Presented for the first time in 2015 with tvOS version 9.0, based on iOS 9. Prior to tvOS, the Apple TV had an operating system called Apple TV Software, which reached version 7.9. Its validity was from 2007 to 2014 with version 7.0 although versions were released until 2022 with final 7.9

It is the only device that has its own TVML language and is one of the options we have to develop apps for Apple TV.

About tvOS (technologies)

- **TVML y TVMLKit**

It allows you to create a client server application, using TVML and Javascript, storing it on a server and rendering the interface with TVMLKit. You can combine SwiftUI or UIKit.

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Sobre tvOS (tecnologías)

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- **SwiftUI**

It allows you to create apps as we know them for iOS. With some limitations and some additions. But in this case using SwiftUI for the interface.

That we don't have tvOS vs iOS

- Almacenamiento en el dispositivo.
- Webkit /Safari Services
- PassKit (Apple Pay and Wallet)
- PDFKit
- Playground Support
- ARkit

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That we don't have tvOS vs iOS

- Core Motion
- Speech
- IOBluetooth
- WidgetKit
- Contacts
- HealthKit



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What do we have tvOS vs iOS?

- TVMLKit
- TVUIKit
- TVServices
- DeviceDiscoveryUI

A circular logo with a blue-to-purple gradient border. Inside the circle, the text "ANDROID NIGHT MADRID" is written in a stylized, blocky font. The word "ANDROID" is on the top line, "NIGHT" is on the middle line, and "MADRID" is on the bottom line. The text is in a light blue color.

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To take into account developing for tvOS

- Focus management
- Applcon y TopSelf
- Specific Xibs (iOS ones cannot be used)
- Text input.
- TabBar

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App Icon y Top Shelf

The icon of a tvOS app is different from an iOS app. They have several layers, specifically 3, Back, Middle and Front. It is for the parallax effect it has when obtaining the focus.

The Top Shelf is the image that is shown in the background when the app icon has the focus. Creating an extension of the app can be made dynamic and that highlights the contents of the app. It is one of the functionalities provided by TVUIKit.

Siri Remote

En tvOS la interacción del usuario con la app es a través del mando. Si queremos capturar esos gestos para personalizar acciones, podemos hacer uso de `UITapGestureRecognizer` y con la propiedad `allowedPressTypes` indicar en qué gestos estamos interesados.

```
let tapGesture = UITapGestureRecognizer(target: self,  
                                     action: #selector(remotePlayButtonTapped))  
let gestures = [NSNumber(value: UIPress.PressType.playPause.rawValue)]  
tapGesture.allowedPressTypes = gestures  
view.addGestureRecognizer(tapGesture)
```

Focus management

The system manages the focus for us but it does it with adjacent elements in the direction we make in the gesture in the control. It's not always what we want / need. To have a more personalized management, the system provides us with some tools.

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Focus management (UIKit)

- **adjustImageWhenAncestorFocused**. Allows UIImageView to respond when an ancestor becomes focused. It allows us to apply a parallax effect to a UIImageView when its superview gets the focus
- **focusedFrameGuide**. It allows us to access the frame of a UIImageView when it has the focus.
- **restoresFocusAfterTransition**. UIViewController property, which allows us to decide if we want the focus to be restored to the element where it was after a transition.
- **canBecomeFocused**. Property of UIView. By default false, overwriting this property we can change that.

Focus management (UIKit)

- **UIFocusGuide**. An object that exposes nonview areas as focusable. It represents an invisible, focusable region that can redirect focus movement to other views.
- **UIFocusEnvironment**. A protocol with a set of methods that define the focus behavior for a branch of the view hierarchy. Conformed by `UIViewController`, `UIView`, `UIWindow` and `UIPresentationController`

Focus management (UIKit)

- **setNeedsFocusUpdate.** Submits a request to the focus engine for a focus update in this environment.
- **updateFocusIfNeeded.** Tells the focus engine to force a focus update immediately.
- **didUpdateFocus(in:with:).** Called immediately after the system updates the focus to a new view.

Focus management (UIKit)

- **UITableView, UICollectionView**

1. **rememberLastFocusedIndexPath**. A Boolean value that indicates whether the table view automatically returns the focus to the cell at the last focused index path
2. **collectionView(_:canFocusItemAt:)**. Asks the delegate whether the item at the specified index path can be focused.
3. **indexPathForPreferredFocusedView(in:)**. Asks the delegate for the table view's index path for the preferred focused view
4. Important if **restoresFocusAfterTransition** on the controller is enabled **rememberLastFocusedIndexPath** can be ignored after a transition.

Delegate function **indexPathForPreferredFocusedView** won't be called the collectionView or tableView to be in **preferredFocusEnvironments**.

The background features a dark blue gradient with two large, flowing, wavy bands in a slightly lighter shade of blue. In the center, there is a circular logo with a blue border. Inside the circle, the text "DESIGNER" is at the top, "MADRID" is at the bottom, and "Demo time" is written in white in the middle.

Demo time

Siri Remote

En tvOS la interacción del usuario con al app es a través del mando. Si queremos capturar esos gestos para personalizar acciones, en SwiftUI tenemos funciones para responder a estas acciones:

```
.onPlayPauseCommand {  
    debugPrint("Play pause button tapped")  
}
```

Focus management (SwiftUI)

- **focusedSection**. View modifier that allows us to get the descendants of this one to focus.
- **CardButtonStyle**. Adds parallax effect to a button or buttons within a view
- **@FocusState**. “Property wrapper” That will allow us to decide which element should be focused.
- **defaultFocus**. Modifier that allows us to indicate which element should get focus
- **Focusable**. Modifier so that a view can get focus
- **IsFocused**. Returns whether the nearest focusable ancestor has focus.

Focus management (SwiftUI)

- **hoverEffect**. Applies a hover effect to this view, (highlight) in tvOS 17 applies a shine and parallax effect
- **@Environment(\.resetFocus)**. Get this environment value and call and call it as a function to force a default focus reevaluation at runtime.
- **focusScope(_ namespace: Namespace.ID)**. Creates a focus scope that SwiftUI uses to limit default focus preferences.

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Demo time

Links of interest

[TV Markup Language \(TVML\)](#)

[Implementing a hybrid TV app with TVMLKit](#)

[TVUIKit](#)

[Creating immersive experiences using a full-screen layout](#)

[Mastering the Living Room With tvOS](#)

[Public Apple Frameworks](#)



Q & A

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